

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - TECHNICIAN CERTIFICATE

CONTINUOUS ASSESSMENT TEST 1 - JANUARY-2026

CST 05102 - OBJECT ORIENTED PROGRAMMING WITH JAVA

SEMESTER: I

YEAR: I

DURATION: 1 Hr 30 Min.

MAX. MARKS: 50

CURATED ANSWERS PROVIDED BY 4LAZIE - SMART IN BRAIN

PART A: SHORT ANSWER QUESTIONS

Q1. Define Java language.

Answer: Java is a high-level, class-based, object-oriented programming language designed to have as few implementation dependencies as possible (Write Once, Run Anywhere).

Smart Tip: Always emphasize "Object-Oriented" and "Platform-independent".

Q2. What is bytecode in java?

Answer: Bytecode is the intermediate, platform-independent code generated by the Java compiler after compiling a .java source file. It is then executed by the Java Virtual Machine (JVM).

Q3. Differentiate concept of polymorphism and abstraction in java.

Answer:

- **Polymorphism:** The ability of an object to take on many forms (e.g., method overloading and overriding).
- **Abstraction:** The process of hiding implementation details and showing only essential features (using abstract classes/interfaces).

Q4. Differentiate between object and class in java.

Answer:

- **Class:** A blueprint or template that defines properties and behaviors common to all objects of a certain kind.
- **Object:** An instance of a class that has specific state and behavior, occupying physical memory.

Q5. Differentiate interface and encapsulation concepts in java.

Answer:

- **Interface:** A blueprint of a class to achieve 100% abstraction and support multiple inheritance.
- **Encapsulation:** The mechanism of wrapping data and code together as a single unit to protect internal state.